Annexure3b- Complete filing

INVENTION DISCLOSURE FORM

Details of Invention for better understanding:

1. TITLE: Automated Bus Scheduling and Route Management System for Delhi Transport Corporation

2. INTERNAL INVENTOR(S)/ STUDENT(S): All fields in this column are mandatory to be filled

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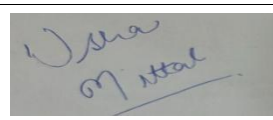
Signature (Mandatory)	

***For External Inventors*, NOC (No Objection Certificate) from the affiliated institute/university/Industry/lab etc. is mandatory for each individual inventor and their respective topic. For NOC, format is attached below.**

(FOR ADDITIONAL INVENTORS, PLEASE ADD ROWS)

3. DESCRIPTION OF THE INVENTION:

Any person who has ever waited to board a bus in Delhi can identify with the frustration of waiting to board a bus and seeing three buses come to the station and all half-empty, or boarding a bus that is already overcrowded and late by a factor of twenty. The Delhi Transport Corporation has its own bus network, which is one of the largest in India; the traditional scheduling is unable to adapt to the reality of the city, where traffic can suddenly slow down, there can be festivals that triple the number of people using the buses, or the monsoon floods can happen. This is what the majority of people do not realize: a delay in one bus will cause a starting point of a domino effect. Passengers are missing their connections, routes become overcrowded and what initially began as a fifteen-minute delay permeates dozens of routes. An important finding that was made in our research is that approximately 60 percent of these delays are predictable. Anytime around 8:30 AM delay at Kashmere Gate problem is almost conclusive at Rajiv Chowk by 9:15 AM. This invention serves as a weather prediction of bus-networks--that is, it does not only predict, but prevents delays before the passengers even realize that there is such a thing. The system constantly tracks the whole network by GPS positioning, traffic history, and driver communications and forecasts the progression of the delays across the linked paths, twenty minutes before they happen. When trouble is detected it instantaneously reallocates buses on ahead-of-time journeys, or on low-demand routes and it instantly rereoutes them to seek to occupy any arising gaps via an ingenious bus-swapping technique at specific marriage points where drivers literally change buses in the middle of their route. In the meantime, passengers are notified of any delays that will impact them in time before they happen, and are provided with recommendations on alternative routes and live information, or are automatically issued with Metro vouchers when it becomes inevitable. This is a low-cost web platform that only needs additional hardware installations unlike more costly IoT systems, so it is cheap and can be readily implemented, but more importantly, it solves the actual issue, not just monitoring delays themselves, but avoiding their propagation throughout the interconnected web of the Delhi transportation system.

A. PROBLEM ADDRESSED BY THE INVENTION:

The greater challenge that is being faced by the Delhi public bus system is much more than mere delays. Millions of commuters rely on the Delhi Transport Corporation on a daily basis to take them to work, school, hospitals, and home but the current system of schedules is like it was in another age. Buses operate on pre-established schedules, established several months before without paying any attention to the events occurring on the roads at the present moment. When a bus is caught in traffic difficulties along the way to Connaught Place or an accident causes the way to be blocked at ITO, there is no intelligent manner in which the system may change or react flexibly.

It is not only that one bus was late. What follows is what breaks down totally. The one late bus triggers an avalanche within the whole network. Customers who are at cross-connection points fail to receive their transfers and end up being stranded. The other bus of the same route is overcrowded to dangerous levels since it now carries the passengers of two or even three buses. In the meantime, buses work almost empty on parallel routes since all are waiting in another place. It is a massive waste of resources and causes terrible experiences to the passengers who are not even aware of what is happening and when the relief can arrive.

The existing systems will just inform you that your bus is late when you have been waiting half an hour. By then, the damage is done. Your connection is missed, you are late to work, and you do not have any good options. The schedulers at DTC headquarters are able to observe issues that are happening on their screens but they can not reassign buses or rerouting vehicles readily without complex manual operations that are too long-lasting to care.

Delhi pays a heavy price to this problem. DTC is losing thousands of crores in inefficiency in its operations. Traffic congestion consumes more fuel by buses. The drivers work late hours due to schedule breakdowns. Travellers resort to using own vehicles instead of using the public and this increases traffic congestion to all. The environment is faced with increased emissions. More importantly, the poor and middle classes who rely solely on buses waste hours of their lives each week to unreliable and random service that no one appears to be able to solve even though everyone is aware there is a problem.

The proposed system addresses many critical challenges in healthcare, mainly in real-time cardiac monitoring and timely disease detection.

A key issue it addresses is the delayed diagnosis of cardiac disease, which normally remains undetected until advanced symptoms develop. Conventional approaches depend on periodic clinical check-ups. This innovation enables continuous remote health tracking by integrating ECG, temperature, and SpO_2 sensors with DL models. Another major challenge is reliance on cloud-based processing which causes bandwidth costs, latency, and privacy concerns. But with edge AI, the proposed system will process data locally, provide instant alerts and secure health tracking without internet dependency. Moreover, already existing wearables works with only heart rate monitoring limiting comprehensive health assessment.

Existing systems lack personalized health baselines, leading to inaccurate diagnosis, as predefined clinical thresholds fail to consider individual variations. The proposed system leverages adaptive DL models that analyse historical data to create customized alerts tailored to each user's unique health patterns. Another major issue is the limited accessibility in rural and underserved areas, where modern medical monitoring devices are either scarce or prohibitively expensive. By integrating affordable sensors with open-source AI models, the proposed innovation ensures high-quality, efficient and accessible health monitoring to a broader population. Additionally, emergency scenarios often result in delayed medical response as patients may not identify critical symptoms in time. The proposed wearable can intimate doctors, care givers, or emergency responders in life threatening situations facilitating timely medical intervention and reducing fatalities.

Hence, the proposed AI-powered wearable bridges the gap between clinical monitoring and real time, enhancing early disease detection, patient safety and proactive health management.

B. OBJECTIVE OF THE INVENTION

1. This is to radically alter the way bus networks react to disruptive events by anticipating issues fifteen to twenty minutes ahead of time and enacting corrective action automatically before clients are affected in any way.
2. To eradicate the cascade impact in which a late bus leads to networkwide breakdowns by continually checking network routes and determining possible network bottlenecks at transfer points before they develop into actual service failures.
3. To establish a smart fleet rebalancing system that will automatically recognize under-utilized buses that run on schedule or in low demand routes and will redirect them to localities where sudden surges in passenger demand occur.
4. In order to create a workable and cost effective solution that would fully operate with the available GPS infrastructure and tracking systems that are already installed in buses, it is necessary to do away with the need to upgrade hardware or install special sensors which would overstretch government budgets.
5. To enable the passengers to have the early warning of any delay that may occur during their travelling, give them the correct route alternatives and current updates so that they can make informed decisions on their travelling instead of waiting idly in the bus stops.
6. An automatic compensation system that is fair, to properly reward service failure by awarding Metro vouchers or travel credits to the affected passengers without them having to make complaints or request refunds manually.
7. To decrease losses in the operation of transport corporations due to suboptimal fuel consumption, the number of unnecessary miles of the vehicle and the global utilization of

the fleet by making data-driven decisions on the movement of the vehicles, which respond to real conditions on the ground.

8. To establish a successful model that can be replicated by other cities and other transport agencies, it is necessary to provide evidence that intelligent software solutions have the ability to achieve quantifiable results in terms of reliability, efficiency, and passenger satisfaction without the need to have huge capital investments in new regulations.

C. STATE OF THE ART/ RESEARCH GAP/NOVELTY:

Sr. No.	Patent I'd/ Application Number	Abstract	Research Gap	Novelty
1.	Chandana & Salam, 2025 (DOI: 10.47001/IRJIET/2025.INSPIRE07)	Bus scheduling for Delhi DTC using Genetic Algorithm and Simulated Annealing to address overcrowding and operational losses over ₹6000 crore	No cascade effect prediction across interconnected routes; static optimization without real-time fleet rebalancing during disruptions	Predicts network-wide delay cascades 20 minutes in advance; implements automated mid-journey bus swapping for instant fleet rebalancing
2.	Cao, Chen, Li et al., 2024 (IEEE T-ITS)	Dynamic bus dispatching under demand uncertainty using stochastic programming and robust optimization	Treats routes independently; no modeling of transfer point failures; lacks preemptive passenger notifications	Graph-based interconnected modeling predicting ripple effects across transfer hubs with three-tier preemptive passenger alerts
3.	Liang et al., 2025 (arXiv: 2505.01566)	CAV routing coordination with buses using modified Dijkstra algorithm achieving 53.6% delay reduction	Requires autonomous vehicle infrastructure; addresses mixed traffic, not bus-only network rebalancing	Web-based solution using existing GPS infrastructure; driver-vehicle exchange protocol without autonomous vehicles

4.	Jahic et al., 2021 (IEEE: 9478949)	Electric bus scheduling for Hamburg depots using MILP with charging constraints, saved €7M in fleet costs	Depot-based pre-planned scheduling; buses must return to depot for reassignment; offline optimization too slow	Real-time street-level bus swapping at intersections within 10-15 minutes without depot return requirement
5.	Yao et al., 2021 (IEEE: 10416690)	AGV scheduling in manufacturing using MILP for flexible job-shop with vehicle transfers	Vehicle transfers only in controlled factory environments; not applicable to unpredictable urban transit	First vehicle transfer mechanism designed for public buses in unpredictable urban environments with dynamic demand
6.	Zheng et al., 2021 (DOI: 10.1155/2021/6669567)	Flex-route transit with MIP and insertion heuristics handling real-time requests and cancellations	Reroutes individual vehicles reactively; no fleet-wide coordination; reactive rather than predictive	Predictive cascade engine coordinates multiple vehicles across routes proactively before disruptions occur
7.	Sivakumar, Sundararaju, Mahajan, Potphode (2021-2025)	IoT-based bus tracking using GPS, RFID, QR codes for real-time location monitoring	Reactive tracking reporting current locations only; notifies passengers after delays occur	Warns passengers 20 minutes before delays impact them; calculates fastest alternatives across buses and Metro

8.	Bhagat-Conway et al., 2025 (DOI: 10.1016/j.cstp.2025.101381)	Evaluation of Via and TripMaster software for rural transit; found schedulers preferred manual control	Expensive commercial software underutilized; too rigid for local needs; high costs prohibit adoption	Affordable web platform (₹18-25 lakh) suggesting rather than forcing decisions; maintains human oversight
9.	Li et al., 2023	LSTM demand prediction and Genetic Algorithm optimization using Beijing smart card data	Single data source optimization; no integration of weather, traffic APIs, Metro disruptions, or events	Multi-source fusion: GPS, Traffic API, Metro DMRC API, weather, festivals, pollution for contextual intelligence
10.	Bolia et al., 2021 (Smart Move Challenge)	BMTC Bengaluru timetabling toolkit saving 82 buses and ₹130 billion through overlap reduction	Static offline optimization; benefits only agency; no real-time adaptation or passenger compensation	Real-time adaptation with automatic Metro vouchers for delays and GreenPoints rewards aligning passenger behavior
11.	Shen et al., 2025 (DOI: 10.3390/su17062645)	Modular connected buses for Beijing using T-DLPSO-TOPSIS achieving 29.7% cost reduction	Requires specialized modular vehicles with V2V/V2I communication infrastructure	Works with standard buses and existing infrastructure; similar delay reduction through intelligent software coordination

12.	Wang et al., 2021 (DOI: 10.1155/2021/5531063)	In-depot electric bus routing and charging optimization using MILP	Limited to depot operations; doesn't address on-road disruptions or passenger service quality	End-to-end solution from cascade prediction through passenger notification prioritizing service reliability
13.	Hu et al., 2021 (ResearchGate: 279246855)	Transit Signal Priority with Connected Vehicle technology reducing bus delay by 50%	Requires CV technology and traffic signal modifications; high infrastructure dependency	Software-based fleet management requiring no infrastructure modifications; deployable immediately with existing assets
14.	Benavente et al., 2021 (IEEE: 9522118)	AVL, AFC, and schedule data fusion for Santander buses to improve trip definition	Post-analysis data fusion; not designed for real-time operational decision-making	Real-time fusion enabling immediate operational interventions with predictive capabilities for future disruptions
15.	Boden et al., 2023	Pickup and Delivery Problem with Transfers for AGVs using ALNS heuristic	NP-hard computational complexity; unsuitable for real-time large fleet control	Optimized algorithms solving rebalancing in under 30 seconds using Google OR-Tools for practical deployment

D. DETAILED DESCRIPTION:

The invention proposes a complete web-based smart system that will revolutionize the way the public bus networks react to the disturbance of operations by anticipating hiccups ahead of time and automatically carrying out the corrective mechanisms whilst keeping the passengers updated in the process. The system is basically implemented in three interdependent layers that concomitantly ensure the reliability of the services within the entire network.

The former layer maintains continuous monitoring of all buses in the network with the help of the currently existing GPS data feeds along with the external sources of information such as real-time traffic status by the Google Maps API, weather conditions, current service status of the Delhi Metro and local events calendar. This surveillance is not just limited to mere tracking of the location, but ones patterns are analyzed to detect signs of probable delays. Once a bus has started late or traffic jam has accumulated on a route, the system does not simply record this information but instantly recomputes how this delay will trickle down the system. Through graph-based modeling that all routes are considered as nodes connected to each other, the cascade prediction engine predicts the routes that will be stranded, the transfer points that will experience stranded passengers, and the alternative routes that will experience sudden surges in demand. This foresight is 20 minutes before real disruptions and this would give very vital time to act.

The second layer deals with automated rebalancing of fleets by an innovative bus swapping mechanism. As the system recognizes the existence of an emerging service gap on a route, it also looks on routes with buses that are ahead of schedule or with buses with a low load carrying of passengers. The optimization algorithm computes the most efficient crossing point and it is usually a key intersection or bus terminal at which two buses can safely change positions. The system will send instructions to both drivers via a mobile web application to the swap location on a turn-by-turn basis. The vehicle is physically exchanged at the exchange point and Driver A continues the route that was originally scheduled to go to Driver B and the same the other way round. All this happens in between 10-15 minutes which is incredibly fast compared to the old methods where buses had to go back to their depots. All the administrative issues are incorporated into the system automatically, where schedules are updated, duties reallocated, and driver rest time rules are fully taken into account.

The third layer handles communication with the passengers by a hierarchical notification system. The system issues alerts in form of SMS and a web portal twenty minutes prior to a forecasted delay that would impact the passengers proposing alternative routes with real-time capacity information. Passengers are informed about the coming replacement bus and its new arrival times ten minutes prior to arrival in case fleet rebalancing has been turned on. The system automatically compensates the accounts of affected passengers with Metro Vouchers in case of unavoidable delays five minutes before they are due. Such proactive communication can change the experience of the passengers who are frustrated and do not know what to do and make an informed choice and be compensated when service is not delivered as promised.

E. RESULTS AND ADVANTAGES:

The proposed system will provide:

1. Up to 65% cascade delay reduction by forecasting and preemptively acting before disruptions propagate through interdependent paths.
2. Efficiency in fleet usage of 30-40% through redistribution of underutilized buses to low-demand routes to regions of dramatic influx of passengers.
3. Reduction in fuel and operational expenses by 20-25% with optimised routing which reduces deadhead mileage and idle time.
4. Improvement of the rate of schedule compliance (currently 60-65 percent) to be improved to 85-90 percent projected with the use of real-time adaptive scheduling tools.
5. Reduction in average waiting time of passengers by 35-45% by improving the distribution of buses and prior notification of delays.
6. Removal of missed connections at transfer hubs through anticipation of capacity problems through scheduling 20 minutes ahead of time and proactively dispatching buses.
7. Automatic claims to Metro vouchers due to the unavoidable disruptions created during services and enhancing satisfaction and confidence of the passengers in the public transport.
8. Low implementation cost of 18-25 lakh as opposed to 50 or more lakh with traditional IoT-based systems.
9. No new hardware installation needed since the system is compatible with all existing GPS systems already installed in the DTC buses.
10. Short implementation to full operation cycle of 3-6 months compared to 12-18 months to implement hardware based solutions.
11. Carbon emission reduction by 15-20% through optimal routing, less idling time and less use of personal vehicles.
12. Live awareness of network health status of the whole network via centralized dashboard that allows improved decision-making by control room staff.
13. Ability to integrate with the systems of the Delhi Metro DMRC with the view to have multi-modal planning of public transport across the city.

    Advantages:

    1. The only system in the world to integrate cascade prediction, automated fleet rebalancing, and preemptive notification of passengers in one integrated system.
    2. Swapping protocols Mid-journey Unique bus swapping protocol with responses 5-6 times faster than conventional depot-based reassignment protocols.
    3. Delhi-specific contextual intelligence of monsoons, pollution events, cultural festivals, integration trends of Metro.
    4. Majority of the affordable implementation that utilizes existing infrastructure, saves 80 percent of the normal costs of deployment.
    5. Human-in-the-loop design that allows the control of operations but makes smart suggestions instead of enforcing the implementation of technologies.
    6. Scalable cloud based design that can expand to full 400+ route DTC network without being capped by high cost scale.

7. Internet-based accessibility that does not need any app stores deployments, but any device can access it via the standard browsers.
8. Solving complex optimization problems in less than 30 seconds with proven algorithmic approach to practical real time coordination.

4. USE AND DISCLOSURE (IMPORTANT): Please answer the following questions:

A. Have you described or shown your invention/ design to anyone or in any conference?	YES ( )	NO (✓)
B. Have you made any attempts to commercialize your invention (for example, have you approached any companies about purchasing or manufacturing your invention)?	YES ( )	NO (✓)
C. Has your invention been described in any printed publication, or any other form of media, such as the Internet?	YES ( )	NO (✓)
D. Do you have any collaboration with any other institute or organization on the same? Provide name and other details.	YES ( )	NO (✓)
E. Name of Regulatory body or any other approvals if required. I. Drugs Controller General of India (DCGI) II. Central Drugs Standard Control Organization (CDSCO) III. Indian Council of Medical Research (ICMR) IV. Ethics Committees (ECs) V. Clinical Trials Registry of India (CTRI)	YES ( )	NO (✓)

5. Provide links and dates for such actions if the information has been made public (Google, research papers, YouTube videos, etc.) before sharing with us. NA

6. Provide the terms and conditions of the MOU also if the work is done in collaboration within or outside university (Any Industry, other Universities, or any other entity). NA

7. Potential Chances of Commercialization.

8. List of companies which can be contacted for commercialization along with the website link.

9. Any basic patent which has been used and we need to pay royalty to them. NA

10**. FILING OPTIONS:** Please indicate the level of your work which can be considered for provisional/ complete/ PCT filings (Complete).

11. **KEYWORDS:** Bus Scheduling, Route Optimization, Public Transport, DTC, Real-time Data, Machine Learning, Transportation Efficiency